Status	Finished
Started	Sunday, 18 October 2026, 9:10 PM
Completed	Sunday, 18 October 2026, 9:15 PM
Duration	5 mins 9 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Sorting the list before searching
d. Checking each element sequentially ✓

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Binary ✓
c. Hash
d. Linear

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search is more efficient than binary search on large lists.
c. Linear search divides the list into halves.
d. Linear search requires the list to be sorted.

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search continues in the left sublist
c. The search continues in the right sublist
d. The search ends successfully ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Sequential search
c. Binary search ✓
d. Linear search

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 94 and 99
c. 89 and 94
d. 90 and 100

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 10 ✓
c. 1000
d. 100

Question **8**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓
b. Selection
c. Insertion
d. Complexity

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Linear search ✓
c. Binary search
d. Jump search

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Binary search algorithm is not efficient when the data elements more than 1500 ✓

d. Must use a sorted array

Question **11**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓

b. 1

c. log n

d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised

b. The last element is returned

c. The first element is returned

d. The search is considered unsuccessful ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓

b. The search continues from the beginning

c. The list is sorted

d. The search stops

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Linear search ✓
c. Interpolation search
d. Hash search

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Binary search
c. Hash search
d. Linear search ✓